Necessity, Sufficiency, and Selectivity: Cross-Task Evaluation of Causal Interventions in Mechanistic Interpretability

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Abstract

Mechanistic interpretability identifies model components that causally mediate specific behaviors, typically through necessity (ablation) and sufficiency (activation patching) interventions. However, these evaluations do not measure whether identified components are *specific* to the target behavior or also affect unrelated behaviors. We introduce cross-task selectivity evaluation as a third criterion for circuit attribution and conduct the first systematic study across three tasks in GPT-2 SMALL: Indirect Object Identification, Subject-Verb Agreement, and Greater-Than comparison. We find that 29–42% of top-ranked components exhibit significant cross-task leakage—they substantially affect behaviors unrelated to their target task. Necessity and sufficiency rankings are highly correlated (Spearman $\rho = 0.76$–0.88) but do not differ significantly in selectivity ($p = 0.52$). The dominant source of leakage is MLP layers, especially MLP_0, which ranks first for all three tasks under both interventions. These findings demonstrate that standard necessity and sufficiency evaluations systematically overstate causal specificity, and that explicit selectivity measurement is essential for valid circuit attribution.

1 Introduction

When mechanistic interpretability researchers claim that a model component "implements" a specific behavior, they typically support this claim with one of two causal interventions: ablating the component degrades the behavior (necessity), or patching in the component's activation recovers it (sufficiency). But neither intervention answers a more fundamental question: does this component affect *only* the targeted behavior?

This question—selectivity—is critical for the downstream applications that motivate mechanistic interpretability. Model editing assumes that modifying a component changes one behavior without affecting others. Safety auditing assumes that circuits identified for dangerous capabilities are specific to those capabilities. Scientific explanations of model internals assume that attributing a behavior to a component is a precise statement about that component's role. If nominally "task-specific" components also mediate unrelated behaviors, all of these applications are undermined.

Despite its importance, selectivity is rarely measured. Mueller et al. [2024] explicitly note that selectivity is "not often explicitly discussed nor measured" in the field. Arora et al. [2024] introduce a selectivity metric for linear features, but no prior work systematically evaluates whether components identified via standard necessity and sufficiency interventions are specific to their target tasks across multiple behaviors.

We address this gap with the first cross-task selectivity evaluation of components identified via activation patching in GPT-2 SMALL. For each of 156 components (144 attention heads and 12 MLP layers), we measure both necessity (noising) and sufficiency (denoising) effects across three tasks spanning distinct linguistic and reasoning domains: Indirect Object Identification (IOI), Subject-

Verb Agreement (SVA), and Greater-Than comparison (GREATER-THAN). We then test each task's top-ranked components on all other tasks to quantify selectivity.

Our results reveal a systematic problem. On average, 29% (necessity) to 42% (sufficiency) of top-15 components for any given task significantly affect at least one unrelated task. The primary culprit is MLP layers: MLP_0 ranks first for *all three tasks* under both interventions, acting as a general-purpose processing stage rather than a task-specific circuit element. Necessity and sufficiency rankings are highly correlated ($\rho = 0.76$–0.88) and do not differ significantly in selectivity ($p = 0.52$, Cohen's $d = 0.115$).

We make the following contributions:

- We conduct the first systematic cross-task selectivity evaluation of model components identified via necessity and sufficiency interventions, quantifying how often "important" components leak across unrelated tasks.
- We compare necessity and sufficiency interventions head-to-head, finding that they produce highly correlated rankings ($\rho = 0.76$–0.88) but do not differ in selectivity.
- We identify MLP layers—especially MLP_0—as the dominant source of cross-task leakage, demonstrating that standard component-level evaluations systematically conflate general-purpose processing with task-specific circuits.
- We propose selectivity as an explicit third evaluation criterion alongside necessity and sufficiency, providing concrete metrics and experimental protocols for future circuit discovery work.

2 Related Work

Causal interventions in mechanistic interpretability. The dominant approach to understanding neural network internals uses causal interventions on model activations. Heimersheim and Nanda [2024] provide a systematic framework connecting intervention direction to causal claims: noising (replacing clean activations with corrupted ones) tests necessity, while denoising (the reverse) tests sufficiency. They show these are not symmetric—AND-structured circuits are fully discoverable via noising, while OR-structured circuits require denoising. Zhang and Nanda [2023] compare corruption strategies and show that symmetric token replacement (STR) avoids the off-distribution artifacts of Gaussian noise. Our work builds on both: we use STR corruption and test both intervention directions, but uniquely evaluate whether identified components are specific to one task or affect many.

Circuit discovery and evaluation. The Indirect Object Identification circuit [Wang et al., 2022] is a landmark result, identifying attention heads that move name information to predict indirect objects. Subsequent work has extended circuit-level analysis to subject-verb agreement, greater-than reasoning, and other tasks. The Mechanistic Interpretability Benchmark [MIB; Mueller et al., 2025] formalizes circuit evaluation with two metrics: Circuit Proportion Recovered (CPR, measuring sufficiency) and Circuit Miss Discovery (CMD, measuring necessity). However, neither MIB nor the original circuit papers measure whether identified components are specific to their target task. We directly address this gap.

Selectivity in interpretability. Mueller et al. [2024] identify four evaluation criteria for causal mediators: sparsity, generality, selectivity, and faithfulness. They note that selectivity—explaining only the target behavior and as little else as possible—is "not often explicitly discussed nor measured." Arora et al. [2024] introduce a selectivity metric for CAUSALGYM, computing the difference between a method's causal efficacy on real tasks versus control tasks. They find that after selectivity adjustment, linear probing outperforms DAS at larger model scales. Unlike CAUSALGYM, which evaluates 1D linear features on linguistic tasks, we evaluate full model components (attention heads and MLP layers) across tasks from different domains.

Feature-level interpretability. An alternative to component-level analysis operates at the feature level. Marks et al. [2024] discover circuits using SAE features and demonstrate selectivity in practice: ablating task-irrelevant features removes gender bias while preserving profession classification. However, Mueller et al. [2025] find that SAE features do not improve over standard neurons for causal variable localization, questioning whether feature decomposition reliably improves selectivity. Sharkey et al. [2025] identify fundamental limitations of sparse dictionary learning, including feature splitting and absorption. Our work complements this line by showing that even at the component level, selectivity failures are systematic and quantifiable.

Causal abstraction. Geiger et al. [2023] provide a formal framework grounding mechanistic interpretability in causal abstraction theory, defining interchange intervention accuracy (IIA) as a measure of counterfactual faithfulness. Bereska and Gavves [2024] survey the field through an AND/OR logic lens, connecting circuit structure to the relative informativeness of necessity and sufficiency tests. Our empirical results align with these theoretical predictions: IOI shows the most divergence between necessity and sufficiency ($\rho = 0.760$), consistent with its known OR-circuit structure with redundant name mover heads.

3 Methodology

3.1 Model and Components

We study GPT-2 SMALL [Radford et al., 2019], a 12-layer transformer with 12 attention heads per layer ($d_{\text{model}} = 768$). We evaluate all 156 components: 144 attention heads (indexed as $\text{H}_{\ell.h}$ for layer ℓ, head h) and 12 MLP layers (indexed as MLP_ℓ). We use TransformerLens for all model access and activation patching.

3.2 Tasks and Datasets

We construct three task datasets, each with 100 clean/corrupted prompt pairs. The tasks span distinct linguistic and reasoning domains to enable meaningful cross-task selectivity evaluation.

Indirect Object Identification (IOI). The model must predict the indirect object in sentences like "When Anna and David went to the store, David gave a ball to __." The correct answer is the name that is *not* the subject of the final clause (here, "Anna").

Subject-Verb Agreement (SVA). The model must predict the correct verb form given a subject with an intervening attractor: "The dog above the books __" → "is" (not "are").

Greater-Than (GREATER-THAN). The model must predict a valid continuation digit for a year comparison: "The war lasted from 1758 to 17__" → any digit ≥ 6.

3.3 Corruption Strategy

Following Zhang and Nanda [2023], we use symmetric token replacement (STR) rather than Gaussian noise to avoid off-distribution artifacts. For IOI, we replace names with different names (changing the correct indirect object). For SVA, we swap singular/plural subjects and attractors (flipping the required verb number). For GREATER-THAN, we change the reference year (shifting valid continuation digits). All tokens are verified as single tokens in GPT-2 SMALL's vocabulary.

3.4 Intervention Procedures

For each component c and task t, we compute two scores:

Necessity (noising). We run the model on clean input x and replace component c's activations with those from the corrupted input $\tilde{x}$. The necessity score measures how much the logit difference drops:

$$\text{Nec}(c, t) = \frac{\Delta_{\text{logit}}(x) - \Delta_{\text{logit}}(x \mid c \leftarrow \tilde{c})}{\Delta_{\text{total}}} \tag{1}$$

where $\Delta_{\text{total}} = \Delta_{\text{logit}}(x) - \Delta_{\text{logit}}(\tilde{x})$ is the total effect of corruption.

Sufficiency (denoising). We run the model on corrupted input $\tilde{x}$ and replace component c's activations with those from the clean input x. The sufficiency score measures how much the logit difference recovers:

$$\text{Suf}(c, t) = \frac{\Delta_{\text{logit}}(\tilde{x} \mid c \leftarrow c^*) - \Delta_{\text{logit}}(\tilde{x})}{\Delta_{\text{total}}} \tag{2}$$

where c^* denotes the clean activation.

Both scores are normalized by Δ_{total} to enable comparison across tasks with different effect magnitudes.

Parameter	Value	Justification
Examples per task	100	Sufficient for stable patching estimates
Batch size	25	Memory-efficient; 4 batches per task
Top-K	15	$\sim$10% of components; standard in MI
Leakage threshold τ	0.05	5% of normalized effect
Random seed	42	Reproducibility

Table 1: Experimental hyperparameters.

Task	Clean Δ_{logit}	Corrupted Δ_{logit}	Total Effect
IOI	1.224	-0.702	1.926
SVA	3.318	-3.311	6.629
GREATER-THAN	2.953	0.523	2.430

Table 2: Logit difference statistics for each task. All tasks show clear signal, with clean inputs producing higher logit differences than corrupted inputs.

3.5 Cross-Task Selectivity

For each task t, we identify the top-K components by absolute effect size ($K = 15$, approximately 10% of all components). We then measure each component's effect on all other tasks.

Selectivity score. For a component c identified as important for task t:

$$\text{Sel}(c, t) = |\text{Effect}(c, t)| - \frac{1}{|T \setminus t|} \sum_{t' \in T \setminus t} |\text{Effect}(c, t')| \tag{3}$$

Positive values indicate the component is more specific to its target task.

Cross-task leakage. A component *leaks* if its maximum absolute effect on any non-target task exceeds a threshold $\tau = 0.05$ (5% of normalized effect). The leakage rate is the fraction of top-K components that leak.

3.6 Statistical Analysis

We compare necessity and sufficiency rankings using Spearman rank correlation. We test whether the two intervention types differ in selectivity using the Mann-Whitney U test (appropriate for non-normal distributions of selectivity scores). We compute bootstrap 95% confidence intervals for the selectivity difference (10,000 resamples) and report Cohen's d for effect size.

3.7 Hyperparameters

table 1 summarizes the experimental configuration.

4 Results

4.1 Task Validation

All three tasks produce clear separation between clean and corrupted inputs (table 2). SVA shows the largest total effect (6.63), while GREATER-THAN shows the smallest (2.43). These differences in scale motivate our use of normalized scores throughout.

4.2 Top Components by Necessity and Sufficiency

table 3 shows the five highest-ranked components for each task under both interventions. Two patterns are immediately apparent. First, MLP_0 ranks first for all three tasks under both necessity and sufficiency, with normalized scores ranging from 0.61 to 1.07. Second, the top-5 lists for each task are largely similar across intervention types, though the exact ordering differs.

	Necessity (Noising)		Sufficiency (Denoising)	
Rank	**Component**	**Score**	**Component**	**Score**
*Indirect Object Identification (*IOI*)*				
1	$\mathbf{MLP_0}$	**0.760**	$\mathbf{MLP_0}$	**1.066**
2	$H_{10.7}$	-0.366	$H_{9.9}$	0.683
3	$H_{11.10}$	-0.256	$H_{10.7}$	-0.513
4	$H_{8.10}$	0.124	$H_{9.6}$	0.317
5	$H_{10.0}$	0.097	$H_{11.10}$	-0.269
*Subject-Verb Agreement (*SVA*)*				
1	$\mathbf{MLP_0}$	**1.034**	$\mathbf{MLP_0}$	**1.033**
2	MLP_8	0.415	MLP_8	0.421
3	MLP_{10}	0.386	MLP_{10}	0.368
4	MLP_{11}	0.282	MLP_{11}	0.271
5	$H_{7.4}$	0.235	$H_{7.4}$	0.227
*Greater-Than (*GREATER-THAN*)*				
1	$\mathbf{MLP_0}$	**0.614**	$\mathbf{MLP_0}$	**0.826**
2	MLP_{10}	0.556	MLP_{10}	0.562
3	$H_{9.1}$	0.453	$H_{9.1}$	0.469
4	MLP_9	0.448	MLP_9	0.429
5	$H_{7.10}$	0.151	$H_{7.10}$	0.179

Table 3: Top-5 components by absolute effect size for each task. MLP_0 (**bold**) ranks first across all tasks and both intervention types. SVA and GREATER-THAN are dominated by MLP layers, while IOI depends more on attention heads.

The task-level composition of the top-5 reveals structural differences. SVA and GREATER-THAN both rely heavily on MLP layers (4/5 and 3/5 of the top-5 are MLPs, respectively), while IOI depends primarily on attention heads (4/5 are heads). This is consistent with the linguistic nature of IOI, which requires attention-mediated name tracking, versus the more general computations underlying verb agreement and numerical comparison.

4.3 Necessity–Sufficiency Correlation

Necessity and sufficiency rankings are highly correlated across all three tasks (table 4). SVA shows the highest correlation ($\rho = 0.883$), suggesting a primarily serial circuit structure where the same components are both necessary and sufficient. IOI shows the lowest ($\rho = 0.760$), consistent with its documented OR-circuit structure containing redundant name mover heads [Wang et al., 2022, Heimersheim and Nanda, 2024]—some heads are sufficient individually but not individually necessary.

A concrete example of divergence: $H_{9.9}$ ranks 2nd by sufficiency for IOI (score 0.683) but is not in the top-5 for necessity. This is a textbook sufficient-but-not-necessary component—patching in

Task	Spearman ρ	p-value
IOI	0.760	1.14×10^{-30}
SVA	**0.883**	1.45×10^{-52}
GREATER-THAN	0.802	3.00×10^{-36}

Table 4: Spearman rank correlation between necessity and sufficiency rankings across all 156 components. All correlations are highly significant ($p < 10^{-30}$). IOI shows the most divergence, consistent with redundant name mover heads.

Task Pair	Necessity Overlap	Sufficiency Overlap
IOI ∩ SVA	3/15 (20%)	5/15 (33%)
IOI ∩ GREATER-THAN	3/15 (20%)	5/15 (33%)
SVA ∩ GREATER-THAN	**7/15 (47%)**	6/15 (40%)

Table 5: Overlap in top-15 components between task pairs. SVA and GREATER-THAN share the most components, reflecting their shared reliance on MLP layers. Sufficiency tends to show equal or higher overlap than necessity.

its clean activation substantially recovers the IOI behavior, but ablating it alone does not degrade performance, because other name mover heads compensate.

4.4 Cross-Task Overlap

table 5 shows the overlap in top-15 components between task pairs. SVA and GREATER-THAN share the most components (47% by necessity, 40% by sufficiency), likely because both rely heavily on MLP-mediated computations. IOI shares only 20–33% with either task, reflecting its distinct attention-head-driven circuit.

4.5 Cross-Task Leakage

Our central finding is that a substantial fraction of top-ranked components leak across tasks. We define leakage as a component's maximum absolute effect on any non-target task exceeding 0.05 (5% of normalized effect).

Across the three tasks, the mean leakage rate is 28.9% for necessity and 42.2% for sufficiency. That is, nearly one in three components identified as necessary for a task, and more than two in five identified as sufficient, also significantly affect at least one unrelated task.

Sufficiency shows higher leakage than necessity (42.2% vs. 28.9%). This may reflect the asymmetry of the two interventions: denoising patches inject a coherent, clean signal into an otherwise corrupted forward pass, and this stronger signal may propagate more broadly through the network. Noising, by contrast, introduces a weaker perturbation that self-repair mechanisms can partially compensate [Heimersheim and Nanda, 2024].

4.6 Selectivity Comparison: Necessity vs. Sufficiency

We test whether necessity and sufficiency differ in selectivity using the selectivity score defined in equation 3. table 6 summarizes the results.

Sufficiency shows marginally higher mean selectivity (0.112 vs. 0.093), but this difference is not statistically significant (Mann-Whitney U, $p = 0.524$). Cohen's $d = 0.115$ indicates a negligible effect size, and the bootstrap 95% confidence interval for the difference ($[-0.046, 0.083]$) contains zero. We conclude that necessity and sufficiency interventions produce similarly (un)selective component rankings.

Statistic	Necessity	Sufficiency
Mean selectivity (± SD)	0.093 ± 0.146	0.112 ± 0.184
Comparison		
Mann-Whitney U	$p = 0.524$	
Cohen's d	0.115	
Bootstrap 95% CI	$[-0.046, 0.083]$	

Table 6: Selectivity comparison between necessity and sufficiency. Sufficiency shows slightly higher mean selectivity, but the difference is not statistically significant. The confidence interval contains zero, and the effect size is negligible.

4.7 MLP_0: A General-Purpose Component

MLP_0 warrants special attention. It ranks first for all three tasks under both interventions, with necessity scores of 0.760 (IOI), 1.034 (SVA), and 0.614 (GREATER-THAN). Its coefficient of variation across tasks ($\text{CV} = 0.22$) is far below that of task-specific components such as $\text{H}_{10.7}$ (which appears in the top-5 only for IOI).

MLP_0 processes the residual stream immediately after the embedding layer. Its high effect across all tasks likely reflects its role in the embedding-to-residual-stream transformation—a general-purpose computation that affects all downstream processing, regardless of the specific task. Including MLP_0 in a task-specific circuit is therefore misleading: its ablation degrades all behaviors, not just the target one.

5 Discussion

5.1 Implications for Circuit Discovery

Our results have direct consequences for how circuits should be identified and reported.

Selectivity as a required criterion. Current practice evaluates circuits on faithfulness (does the circuit reproduce the behavior?) and completeness (does removing the circuit degrade it?). Our finding that 29–42% of top-ranked components leak across tasks shows that these criteria are insufficient. A component can score highly on both necessity and sufficiency for a given task while also affecting unrelated tasks. We argue that selectivity should be reported alongside faithfulness and completeness in all circuit analyses.

MLP layers need special treatment. The dominance of MLP_0 across all tasks reveals a category error in standard component-level analysis. Early MLP layers perform general-purpose transformations (embedding processing, residual stream initialization) that are necessary for *all* model behaviors. Including them in task-specific circuits conflates infrastructure with mechanism. We recommend either excluding early MLP layers from circuit analyses or explicitly distinguishing general-purpose components from task-specific ones.

Implications for model editing. If a practitioner edits MLP_0 to change the model's behavior on IOI, our results predict that SVA and GREATER-THAN performance will also be affected. More broadly, any model editing technique that targets components with low selectivity risks unintended side effects. Cross-task selectivity evaluation should precede any targeted intervention on model internals.

5.2 Why Sufficiency and Necessity Do Not Differ in Selectivity

We hypothesized that sufficiency (denoising) would produce more selective component rankings than necessity (noising), reasoning that sufficiency tests a more targeted causal claim. This hypothesis was not supported ($p = 0.524$). Two factors may explain this null result.

First, at the component level, both interventions modify the same activations—they differ only in the direction of replacement (clean $\rightarrow$ corrupted vs. corrupted $\rightarrow$ clean). A component that mediates multiple tasks will show effects under both interventions, regardless of direction.

Second, the high correlation between necessity and sufficiency rankings ($\rho = 0.76$–0.88) means that both methods largely identify the same components. If the same components are selected, their selectivity profiles will naturally be similar. A meaningful difference might emerge with finer-grained mediators (e.g., SAE features or learned subspaces) where the mapping between necessity and sufficiency is less tight.

5.3 Task Similarity Drives Overlap

The pattern of cross-task overlap is informative. SVA and GREATER-THAN share 40–47% of their top components, while IOI shares only 20–33% with either. This is not random: SVA and GREATER-THAN both involve syntactic/numerical processing that relies on MLP layers, while IOI depends on attention-head-mediated name tracking. The degree of component overlap thus reflects genuine computational similarity between tasks, not just noise from general-purpose components.

This observation suggests a constructive use of cross-task evaluation: the pattern of shared and distinct components across tasks could serve as a *fingerprint* of computational similarity, revealing which tasks share mechanisms even when their surface descriptions differ.

5.4 Limitations

Single model. Our experiments use only GPT-2 SMALL. Larger models with greater capacity may develop more specialized circuits with higher selectivity. Conversely, they may also develop more complex MLP computations that leak more broadly.

Three tasks. Three tasks provide limited statistical power for cross-task generalization claims. The leakage rates we report (29–42%) are specific to this task set. Different tasks—or more tasks—could yield different rates.

Component granularity. We patch entire attention heads and full MLP layers. Finer-grained analysis at the level of individual neurons, SAE features [Marks et al., 2024], or DAS-identified subspaces [Geiger et al., 2023] might reveal more selective units within these components.

Single-component interventions. We test components individually. Multi-component interventions would reveal circuit-level interactions, such as whether a set of components is jointly selective even if individual members are not.

Self-repair confound. Necessity scores may be underestimated due to backup mechanisms that compensate for ablated components [Heimersheim and Nanda, 2024]. This could inflate the apparent leakage rate for necessity if compensation partially masks a component's true contribution to non-target tasks as well.

6 Conclusion

We conducted the first systematic cross-task selectivity evaluation of model components identified via necessity and sufficiency interventions in GPT-2 SMALL. Our findings are clear: standard causal interventions systematically overstate the specificity of identified components. Across three tasks, 29–42% of top-ranked components significantly affect unrelated behaviors, with MLP layers—especially MLP_0—acting as general-purpose processing stages rather than task-specific circuit elements. Necessity and sufficiency interventions produce highly correlated rankings ($\rho = 0.76$–0.88) and do not differ significantly in selectivity.

These results argue for a simple change in practice: measure selectivity explicitly. A component being necessary or sufficient for a behavior does not mean it is *specific* to that behavior, and failing to check this distinction undermines circuit-level explanations, model editing, and safety auditing. We propose selectivity as a third evaluation criterion alongside necessity and sufficiency and provide concrete metrics and protocols for its measurement.

Future work should extend this analysis to larger models, finer-grained components (e.g., SAE features), and multi-component interventions. A particularly promising direction is using cross-task selectivity patterns as a computational similarity fingerprint—a way to discover which tasks share mechanisms based on which components they share.

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